

Exercício Prático 06

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Parte 1 - Responda

1. O que é um arquivo fonte?

A. um arquivo de texto que contém instruções de linguagem de programação.

2. O que é um registrador?

B. uma parte do processador que possui um padrão de bits.

3. Qual o caracter que, na linguagem assembly do SPIM, inicia um comentário?

A. #.

4. Quantos bits há em cada instrução de máquina MIPS?

C. 32.

5. O que é o contador de programa?

D. parte do processador que contém o endereço da próxima instrução de máquina para ser obtida.

6. Ao executarmos uma instrução, quanto será adicionado ao contador de programa?

C. 4.

7. O que é uma diretiva, tal como a diretiva .text?

D. uma declaração que diz o montador algo sobre o que o programador quer, mas não corresponde diretamente a uma instrução de máquina.

8. O que é um endereço simbólico?

D. um nome usado no código-fonte em linguagem assembly para um local na memória.

9. Em qual endereço o simulador SPIM coloca a primeira instrução de máquina quando ele está sendo executado?

B. 0x00400000.

10. Algumas instruções de máquina possuem uma constante como um dos operandos. Como é chamado tal operando?

A. operando imediato.

11. Como é chamada uma operação lógica executada entre bits de cada coluna dos operandos para produzir um bit de resultado para cada coluna?

B. operação bitwise.

12. Quando uma operação é de fato executada, como estão os operandos na ALU?

D. Cada um dos registradores deve possuir 32 bit.

13. Dezesesseis bits de dados de uma instrução de ori são usados como um operando imediato. Durante execução, o que deve ser feito primeiro?

B. Os dados são estendidos em zero à esquerda por 16 bits.

14. Qual das instruções seguintes armazenam no registrador \$5 um padrão de bits que representa positivo 48?

A. ori \$5,\$0,0x48.

15. A instrução de ori pode armazenar o complemento de dois de um número em um registrador?

A. Não.

16. Qual das instruções seguintes limpa todos os bits no registrador \$8 com exceção do byte de baixa ordem que fica inalterado?

D. andi \$8,\$8,0xFF

17. Qual é o resultado de um ou exclusivo de padrão sobre ele mesmo?

A. Todos os bits em zero.

18. Todas as instruções de máquina têm os mesmos campos?

A. Não. Diferentes de instruções de máquina possuem campos diferentes.

Parte 2 - Implementar em MIPS/MARS os seguintes programas (usando apenas as instruções indicadas)

Programa 1:

The screenshot shows the MARS MIPS assembler interface. The main window displays the assembly code for 'mips1.asm'. The code is as follows:

```
1 #programa 1 (add, addi, sub, lógicas)
2 #a =2;
3 #b =3;
4 #c =4;
5 #d =5;
6 #x = (a+b) - (c+d);
7 #y = a - b + x;
8 #b = x - y;
9
10 # a= $s0; b= $s1; c= $s2; d= $s3; x= $s4; y= $s5;
11
12 #inicio
13 .text
14 .globl main
15 main:
16     ori $s0, $zero, 2 # a = 2
17     ori $s1, $zero, 3 # b = 3
18     ori $s2, $zero, 4 # c = 4
19     ori $s3, $zero, 5 # d = 5
20     add $t0, $s0, $s1 # t0 = a + b
21     add $t1, $s2, $s3 # t1 = c + d
22     sub $s4, $t0, $t1 # x = t0 - t1
23     sub $t3, $s0, $s1 # t3 = a - b
24     add $s5, $t3, $s4 # y = t3 + x
25     sub $s1, $s4, $s5 # b = x - y
26 #fim
```

The status bar indicates 'Line: 11 Column: 1' and 'Show Line Numbers' is checked. Below the code window, the 'Mars Messages' panel shows the following messages:

- Step: execution terminated due to null instruction.
- Assemble: assembling C:\Users\thigo\Downloads\mips1.asm
- Assemble: operation completed successfully.

The screenshot shows the MARS MIPS simulator interface. The main window displays the execution of the program. The 'Text Segment' window shows the assembly code with addresses and values. The 'Data Segment' window shows the memory layout. The 'Registers' window shows the state of the registers.

Text Segment:

Bkpt	Address	Code	Basic	Source
	0x00400000	ori \$16,\$0,0x00000002	16:	ori \$a0, \$zero, 2 # a = 2
	0x00400004	ori \$17,\$0,0x00000003	17:	ori \$a1, \$zero, 3 # b = 3
	0x00400008	ori \$18,\$0,0x00000004	18:	ori \$a2, \$zero, 4 # c = 4
	0x0040000c	ori \$19,\$0,0x00000005	19:	ori \$a3, \$zero, 5 # d = 5
	0x00400010	add \$8,\$0,\$17	21:	add \$t0, \$s0, \$s1 # t0 = a + b
	0x00400014	add \$5,\$18,\$19	22:	add \$t1, \$s2, \$s3 # t1 = c + d
	0x00400018	sub \$20,\$8,\$9	23:	sub \$s4, \$t0, \$t1 # x = t0 - t1
	0x0040001c	sub \$11,\$16,\$17	25:	sub \$t3, \$s0, \$s1 # t3 = a - b
	0x00400020	add \$21,\$11,\$20	26:	add \$s5, \$t3, \$s4 # y = t3 + x
	0x00400024	sub \$17,\$20,\$21	27:	sub \$s1, \$s4, \$s5 # b = x - y

Data Segment:

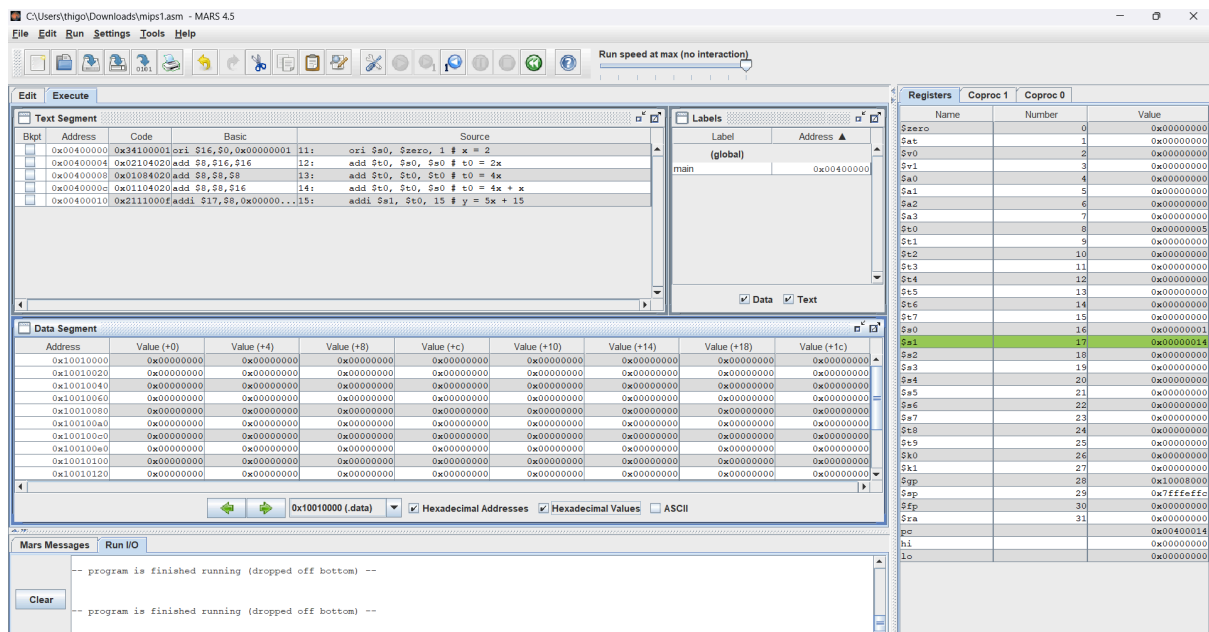
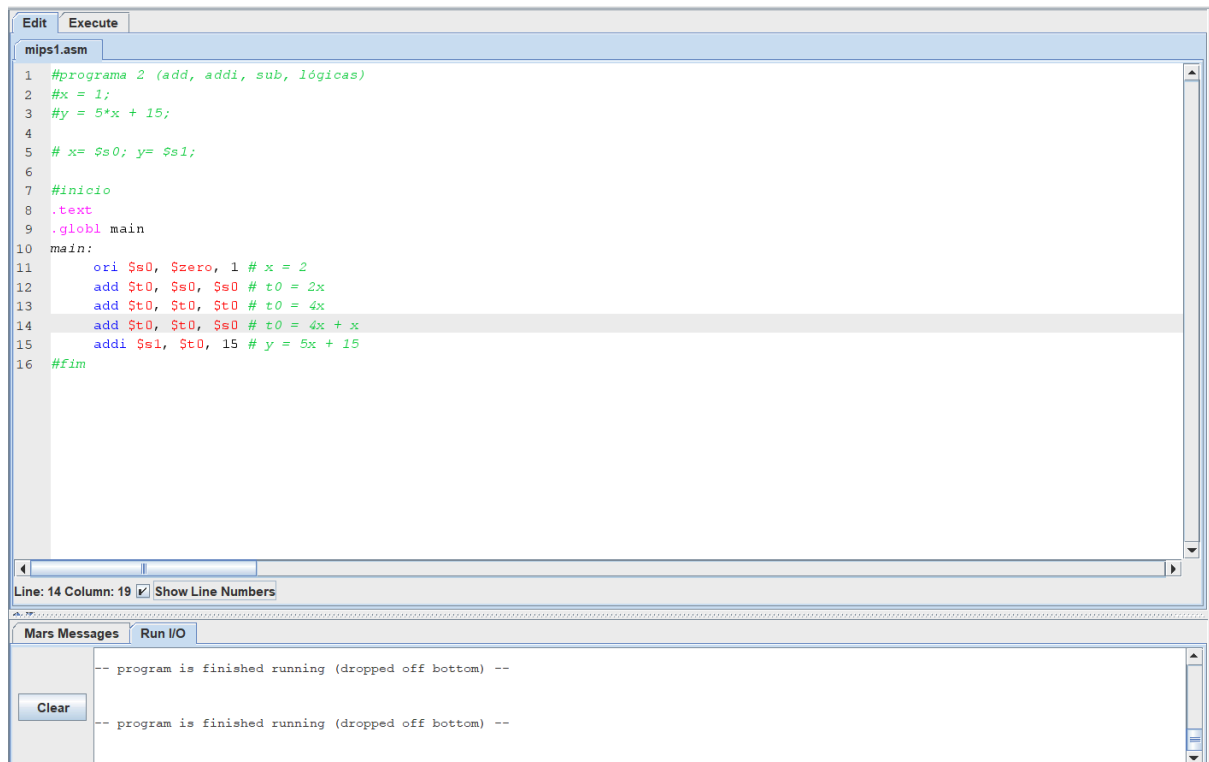
Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers:

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x00000005
\$t1	9	0x00000009
\$t2	10	0x00000000
\$t3	11	0xffffffff
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00000002
\$s1	17	0x00000003
\$s2	18	0x00000004
\$s3	19	0x00000005
\$s4	20	0xffffffff
\$s5	21	0xffffffff
\$s6	22	0x00000000
\$s7	23	0x00000000
\$s8	24	0x00000000
\$s9	25	0x00000000
\$k0	26	0x00000000
\$k1	27	0x00000000
\$gp	28	0x10008000
\$fp	29	0xffffffff
\$fp	30	0x00000000
\$ra	31	0x00000000
\$pc		0x00400028
\$hi		0x00000000
\$lo		0x00000000

The 'Mars Messages' panel shows the message: '-- program is finished running (dropped off bottom) --'.

Programa 2:



Programa 3:

```

#programa 3 (add, addi, sub, lógicas)
#x = 3;
#y = 4 ;
#z = ( 15*x + 67*y)*4
# x= $s0; y= $s1; z= $s2;

#inicio

.text
.globl main
main:

    ori $s0, $zero, 3 # x = 3
    ori $s1, $zero, 4 # y = 4
    add $t0, $s0, $s0 # t0 = 2x
    add $t0, $t0, $t0 # t0 = 4x
    add $t0, $t0, $t0 # t0 = 8x
    add $t0, $t0, $t0 # t0 = 16x
    sub $t0, $t0, $s0 # t0 = 16x - x = 15x
    add $t1, $s1, $s1 # t1 = 2y
    add $t1, $t1, $t1 # t1 = 4y
    add $t1, $t1, $t1 # t1 = 8y
    add $t1, $t1, $t1 # t1 = 16y
    add $t1, $t1, $t1 # t1 = 32y
    add $t1, $t1, $t1 # t1 = 64y
    add $t1, $t1, $s1 # t1 = 65y
    add $t1, $t1, $s1 # t1 = 66y
    add $t1, $t1, $s1 # t1 = 67y
    add $t2, $t0, $t1 # t2 = t0 + t1
    add $t2, $t2, $t2 # t2 = 2*t2
    add $t2, $t2, $t2 # t2 = 4*t2
    add $s2, $zero, $t2 # z = (15x + 67y)* 4

#fim

```

C:\Users\thigo\Downloads\mips1.asm - MARS 4.5

File Edit Run Settings Tools Help

Run speed at max (no interaction)

Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x00000020
\$t1	9	0x0000010c
\$t2	10	0x000004e4
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$a0	16	0x00000003
\$a1	17	0x00000004
\$a2	18	0x000004e4
\$a3	19	0x00000000
\$a4	20	0x00000000
\$a5	21	0x00000000
\$a6	22	0x00000000
\$a7	23	0x00000000
\$t8	24	0x00000000
\$t9	25	0x00000000
\$t0	26	0x00000000
\$t1	27	0x00000000
\$t2	28	0x10008000
\$t3	29	0x7fffffc0
\$t4	30	0x00000000
\$t5	31	0x00000000
\$f0		0x00400000
\$f1		0x00000000
\$f2		0x00000000

Mars Messages Run I/O

-- program is finished running (dropped off bottom) --

Clear

-- program is finished running (dropped off bottom) --

Programa 4:

```
#programa 4
#x = 3;
#y = 4 ;
#z = ( 15*x + 67*y)*4

# x= $s0; y= $s1; z= $s2;

#inicio
.text
.globl main
main:
    ori $s0, $zero, 3 # x = 3
    ori $s1, $zero, 4 # y = 4

    sll $t0, $s0, 4 # t0 = 16x
    sub $t0, $t0, $s0 # t0 = 15x

    sll $t1, $s1, 6 # t1 = 64x
    add $t1, $t1, $s1 # t1 = 65x
    add $t1, $t1, $s1 # t1 = 66x
    add $t1, $t1, $s1 # t1 = 67x

    add $t2, $t0, $t1 # t2 = t0 + t1
    sll $t2, $t2, 2 # t2 = 4*t2
    add $s2, $zero, $t2 # z = t2

#fim
```

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File Edit Run Settings Tools Help

Run speed at max (no interaction)

Execute

Text Segment

Addr	Code	Basic	Source
0x00400000	0x34100003	ori \$t0, \$zero, 3	12: ori \$t0, \$zero, 3 # x = 3
0x00400004	0x34110004	ori \$s1, \$zero, 4	13: ori \$s1, \$zero, 4 # y = 4
0x00400008	0x00104100	sll \$t0, \$s0, 4	15: sll \$t0, \$s0, 4 # t0 = 16x
0x0040000c	0x01104022	sub \$t0, \$t0, \$s0	16: sub \$t0, \$t0, \$s0 # t0 = 15x
0x00400010	0x00114580	sll \$t1, \$s1, 6	18: sll \$t1, \$s1, 6 # t1 = 64x
0x00400014	0x01314820	add \$t1, \$t1, \$s1	19: add \$t1, \$t1, \$s1 # t1 = 65x
0x00400018	0x01314820	add \$t1, \$t1, \$s1	20: add \$t1, \$t1, \$s1 # t1 = 66x
0x0040001c	0x01314820	add \$t1, \$t1, \$s1	21: add \$t1, \$t1, \$s1 # t1 = 67x
0x00400020	0x01055020	add \$t2, \$t0, \$t1	23: add \$t2, \$t0, \$t1 # t2 = t0 + t1
0x00400024	0x000a5080	sll \$t2, \$t2, 2	24: sll \$t2, \$t2, 2 # t2 = 4*t2
0x00400028	0x000a9020	add \$s2, \$zero, \$t2	25: add \$s2, \$zero, \$t2 # z = t2

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Registers

Name	Coproc 1	Coproc 0	Value
\$zero			0x00000000
\$at			0x00000000
\$v0			0x00000000
\$v1			0x00000000
\$a0			0x00000000
\$a1			0x00000000
\$a2			0x00000000
\$a3			0x00000000
\$s0			0x00000000
\$s1			0x00000000
\$s2			0x00000000
\$t0			0x00000000
\$t1			0x00000000
\$t2			0x00000000
\$t3			0x00000000
\$t4			0x00000000
\$t5			0x00000000
\$t6			0x00000000
\$t7			0x00000000
\$s0			0x00000000
\$s1			0x00000000
\$s2			0x00000000
\$a3			0x00000000
\$a4			0x00000000
\$a5			0x00000000
\$a6			0x00000000
\$a7			0x00000000
\$a8			0x00000000
\$a9			0x00000000
\$k0			0x00000000
\$k1			0x00000000
\$gp			0x00000000
\$fp			0x00000000
\$ra			0x00000000
\$pc			0x00000000
\$hi			0x00000000
\$lo			0x00000000

Mars Messages

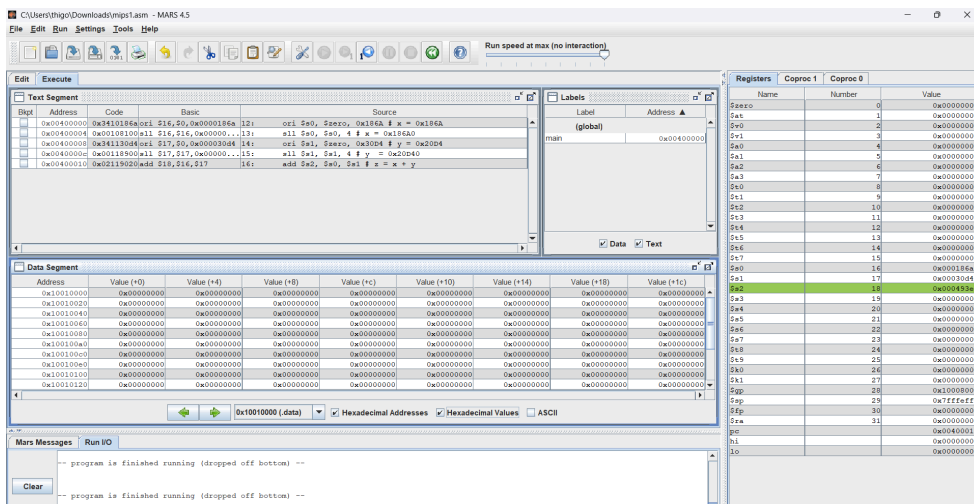
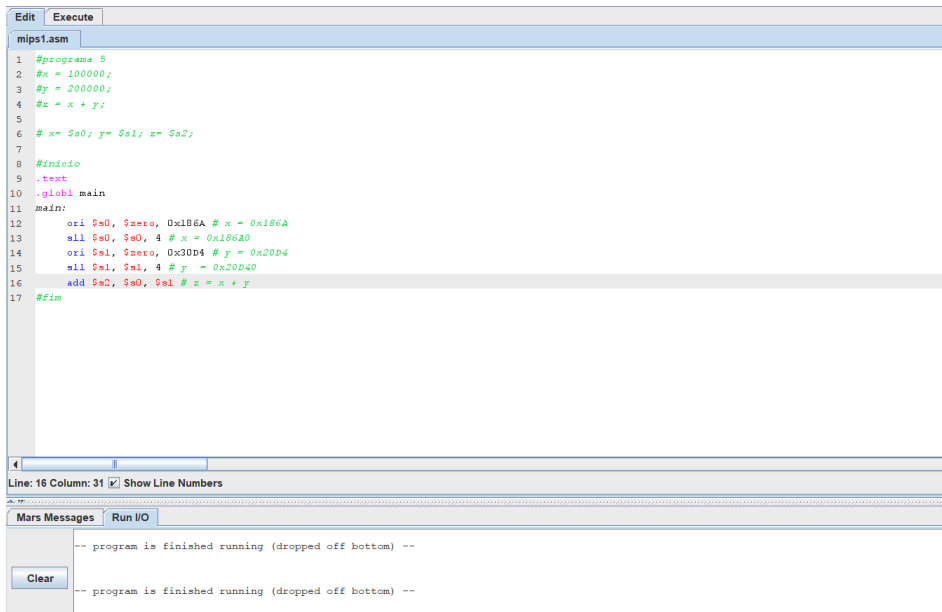
Run I/O

Clear

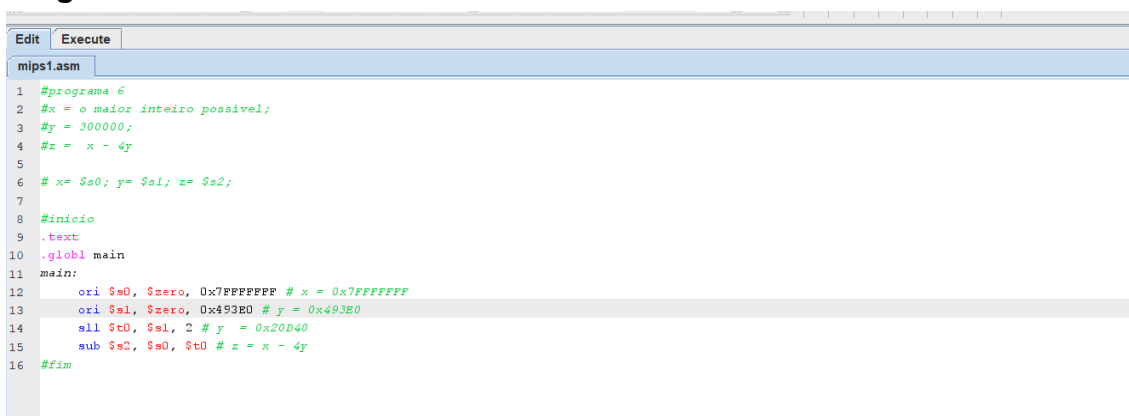
-- program is finished running (dropped off bottom) --

-- program is finished running (dropped off bottom) --

Programa 5:



Programa 6:



C:\Users\thigo\Downloads\mips1.asm - MARS 4.5

File Edit Run Settings Tools Help

Run speed at max (no interaction)

Edit Execute

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x3e07ffff	lui \$1,0x00007fff	12: ori \$a0, \$zero, 0x7fffffff # x = 0x7fffffff
	0x00400004	0x3421ffff	ori \$1,\$1,0x0000ffff	
	0x00400008	0x00018025	or \$16,\$0,\$1	
	0x0040000c	0x3c010004	lui \$1,0x00000004	13: ori \$a1, \$zero, 0x493e0 # y = 0x493e0
	0x00400010	0x342193e0	ori \$1,\$1,0x000093e0	
	0x00400014	0x00018225	or \$17,\$0,\$1	
	0x00400018	0x00114080	sll \$8,\$17,0x00000002	14: sll \$t0, \$s1, 2 # y = 0x20040
	0x0040001c	0x02089022	sub \$18,\$16,\$8	15: sub \$a2, \$a0, \$t0 # z = x - 4y

Labels

Label	Address
(global)	
main	0x00400000

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Mars Messages Run I/O

-- program is finished running (dropped off bottom) --

Clear

-- program is finished running (dropped off bottom) --

Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x0000493e0
\$v0	2	0x000000000
\$v1	3	0x000000000
\$a0	4	0x000000000
\$a1	5	0x000000000
\$a2	6	0x000000000
\$a3	7	0x000000000
\$t0	8	0x000000000
\$t1	9	0x000000000
\$t2	10	0x000000000
\$t3	11	0x000000000
\$t4	12	0x000000000
\$t5	13	0x000000000
\$t6	14	0x000000000
\$t7	15	0x000000000
\$a0	16	0x7fffffff
\$a1	17	0x0000493e0
\$a2	18	0x7feab07f
\$a3	19	0x000000000
\$a4	20	0x000000000
\$a5	21	0x000000000
\$a6	22	0x000000000
\$a7	23	0x000000000
\$t8	24	0x000000000
\$t9	25	0x000000000
\$k0	26	0x000000000
\$k1	27	0x000000000
\$gp	28	0x10008000
\$sp	29	0x7fffffc0
\$fp	30	0x000000000
\$ra	31	0x000000000
pc		0x00400020
hi		0x00000000
lo		0x00000000

Programa 7:

C:\Users\thigo\Downloads\mips1.asm - MARS 4.5

File Edit Run Settings Tools Help

Run speed at max (no interaction)

Edit Execute

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x34080001	ori \$8,\$0,0x00000001	13: ori \$8, \$zero, 0x01 # \$8 = 0x00000001
	0x00400004	0x3508ffff	ori \$8,\$8,0x0000ffff	14: ori \$8, \$8, 0xffff # \$8 = 0x0000ffff
	0x00400008	0x00084000	sll \$8,\$8,16 # \$8 = 0x00000000	
	0x0040000c	0x3508ffff	ori \$8,\$8,0x0000ffff	16: ori \$8, \$8, 0xffff # 0xffff

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Mars Messages Run I/O

step: execution terminated due to null instruction.

Assemble: assembling C:\Users\thigo\Downloads\mips1.asm

Assemble: operation completed successfully.

Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0x000000000
\$at	1	0x000000000
\$v0	2	0x000000000
\$v1	3	0x000000000
\$a0	4	0x000000000
\$a1	5	0x000000000
\$a2	6	0x000000000
\$a3	7	0x000000000
\$t0	8	0x000000000
\$t1	9	0x000000000
\$t2	10	0x000000000
\$t3	11	0x000000000
\$t4	12	0x000000000
\$t5	13	0x000000000
\$t6	14	0x000000000
\$t7	15	0x000000000
\$a0	16	0x000000000
\$a1	17	0x000000000
\$a2	18	0x000000000
\$a3	19	0x000000000
\$a4	20	0x000000000
\$a5	21	0x000000000
\$a6	22	0x000000000
\$a7	23	0x000000000
\$t8	24	0x000000000
\$t9	25	0x000000000
\$k0	26	0x000000000
\$k1	27	0x000000000
\$gp	28	0x10008000
\$sp	29	0x7fffffc0
\$fp	30	0x000000000
\$ra	31	0x000000000
pc		0x00400000
hi		0x00000000
lo		0x00000000



C:\Users\thigo\Downloads\mips1.asm - MARS 4.5

File Edit Run Settings Tools Help

Run speed at max (no interaction)

Edit Execute

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x34081001	ori \$8,\$0,4097	23: ori \$t0, \$zero, 0x1001 # t0 = 0x00001001
	0x00400004	0x00084400	all \$9,\$9,16	24: all \$t0, \$t0, 16 # t0 = 0x10010000
	0x00400008	0x8d090000	lw \$9,\$0(\$8)	26: lw \$t1, 0 (\$t0) # t1 = memoria[0]
	0x0040000c	0x8d0a0004	lw \$10,\$4(\$8)	27: lw \$t2, 4 (\$t0) # t2 = memoria[1]
	0x00400010	0x8d0b0008	lw \$11,\$8(\$8)	28: lw \$t3, 8 (\$t0) # t3 = memoria[2]
	0x00400014	0x8d0c000c	lw \$12,\$12(\$8)	29: lw \$t4, 12 (\$t0) # t4 = memoria[3]
	0x00400018	0x12a68200	add \$19,\$9,\$10	31: add \$t5, \$t1, \$t2 # t5 = t1 + t2
	0x0040001c	0x016c7020	add \$14,\$11,\$12	32: add \$t6, \$t3, \$t4 # t6 = t3 + t4
	0x00400020	0x01ae7820	add \$19,\$11,\$14	33: add \$t7, \$t5, \$t6 # t7 = t5 + t6
	0x00400024	0xad0f0010	sw \$15,\$16(\$8)	35: sw \$t7, 16 (\$t0) # memoria[4]

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	15	25	13	17	70	0	0	0
0x10010020	0	0	0	0	0	0	0	0
0x10010040	0	0	0	0	0	0	0	0
0x10010060	0	0	0	0	0	0	0	0
0x10010080	0	0	0	0	0	0	0	0
0x100100a0	0	0	0	0	0	0	0	0
0x100100c0	0	0	0	0	0	0	0	0
0x100100e0	0	0	0	0	0	0	0	0
0x10010100	0	0	0	0	0	0	0	0
0x10010120	0	0	0	0	0	0	0	0

Mars Messages Run IO

Clear

-- program is finished running (dropped off bottom) --

-- program is finished running (dropped off bottom) --

Registers

Name	Number	Value
\$zero	0	0
\$at	1	0
\$v0	2	0
\$v1	3	0
\$a0	4	0
\$a1	5	0
\$a2	6	0
\$a3	7	0
\$t0	8	268500954
\$t1	9	15
\$t2	10	25
\$t3	11	13
\$t4	12	17
\$t5	13	40
\$t6	14	36
\$t7	15	70
\$a0	16	0
\$a1	17	0
\$a2	18	0
\$a3	19	0
\$a4	20	0
\$a5	21	0
\$a6	22	0
\$a7	23	0
\$t8	24	0
\$t9	25	0
\$k0	26	0
\$k1	27	0
\$gp	28	268468224
\$fp	29	2147475544
\$ra	30	0
\$ra	31	4194344
\$lo		0

Programa 10:

Edit Execute

mips1.asm

```
11 # x= $s0; z= $s1; y= $s2;
12 #inicio
13 .data
14 x: .word 5
15 z: .word 7
16 y: .word 0
17
18 .text
19 .globl main
20 main:
21     ori $t0, $zero, 0x1001 # t0 = 0x00001001
22     all $t0, $t0, 16 # t0 = 0x10010000
23
24     lw $s0, 0 ($t0) # t0 = memoria[0]
25     lw $s1, 4 ($t0) # s1 = memoria[1]
26
27     all $t1, $s0, 7 # t1 = 128x
28     sub $t1, $t1, $s0 # t1 = 127x
29
30     all $t2, $s1, 6 # t2 = 64x
31     add $t2, $t2, $s1 # t2 = 65x
32
33     sub $t3, $t1, $t2 # t3 = t1 - t2
34     addi $s2, $t3, 1 # s2 = t3 + 1
35
36     sw $s2, 8 ($t0) # memoria[2]
37     #fim
```

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File Edit Run Settings Tools Help

Run speed at max (no interaction)

Edit Execute

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x34081001	ori \$8,\$0,4097	23: ori \$t0, \$zero, 0x1001 # t0 = 0x00001001
	0x00400004	0x00084400	all \$9,\$9,16	24: all \$t0, \$t0, 16 # t0 = 0x10010000
	0x00400008	0x8d100000	lw \$16,0x00000000(\$8)	25: lw \$s0, 0 (\$t0) # t0 = memoria[0]
	0x0040000c	0x8d110004	lw \$17,0x00000004(\$8)	26: lw \$s1, 4 (\$t0) # s1 = memoria[1]
	0x00400010	0x00104820	all \$9,\$9,16	28: all \$t1, \$s0, 7 # t1 = 128x
	0x00400014	0x00104822	sub \$9,\$9,16	29: sub \$t1, \$t1, \$s0 # t1 = 127x
	0x00400018	0x00115180	all \$10,\$10,6	30: all \$t2, \$s1, 6 # t2 = 64x
	0x0040001c	0x00115182	add \$10,\$10,6	31: add \$t2, \$t2, \$s1 # t2 = 65x
	0x00400020	0x0012a682	sub \$11,\$11,\$10	33: sub \$t3, \$t1, \$t2 # t3 = t1 - t2
	0x00400024	0x01720001	addi \$18,\$11,1	34: addi \$s2, \$t3, 1 # s2 = t3 + 1
	0x00400028	0xad120008	sw \$18,0x00000008(\$8)	37: sw \$s2, 8 (\$t0) # memoria[2]

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000005	0x00000007	0x00000005	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Mars Messages Run IO

Clear

-- program is finished running (dropped off bottom) --

-- program is finished running (dropped off bottom) --

Registers

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x10010000
\$t1	9	0x0000007b
\$t2	10	0x00000017
\$t3	11	0x00000004
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000007
\$a0	16	0x00000000
\$a1	17	0x00000000
\$a2	18	0x00000005
\$a3	19	0x00000007
\$a4	20	0x00000000
\$a5	21	0x00000000
\$a6	22	0x00000000
\$a7	23	0x00000000
\$t8	24	0x00000000
\$t9	25	0x00000000
\$k0	26	0x00000000
\$k1	27	0x00000000
\$gp	28	0x10000000
\$fp	29	0xffffffff
\$ra	30	0x00000000
\$ra	31	0x00000000
\$lo		0x00000000

Programa 11:

The screenshot displays the MARS MIPS simulator interface. The top window shows the assembly code for `mips1.asm`, which includes comments in Portuguese and assembly instructions. The code initializes registers `$x0`, `$x1`, and `$x2`, then performs a series of operations on registers `$t0`, `$t1`, and `$t2` using instructions like `ori`, `sll`, `lw`, `sub`, and `add`. The bottom window shows the MARS interface with the assembly code loaded. The `Text Segment` and `Data Segment` are visible, showing memory addresses and values. The `Registers` window on the right shows the state of the MIPS registers, with `$zero` at 0 and `$t0` at 0x00001001. The `Mars Messages` window at the bottom shows the message "program is finished running (dropped off bottom)".

```
1  #!: .word 0  # esse valor deverá ser sobrescrito após a execução do programa.
2
3  # x= $s0; z= $s1; y= $s2;
4  #inicio
5  .data
6  x: .word 100000
7  z: .word 200000
8  y: .word 0  # esse valor deverá ser sobrescrito após a execução do programa.
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```

Programa 12:

```
mips1.asm
9  #k = MEM [ MEM [MEM [ x ] ] ].
10 #Crie um programa que implemente a estrutura de dados acima, leia o valor de K, o multiplique por
11 #2 e o reescreva no local correto conhecendo-se apenas o valor de x.
12
13 # k= $s0;
14 #inicio
15 .data
16 x: .word p2          # Terceiro ponteiro que aponta para 'p2' (***x)
17 p2: .word p1          # Segundo ponteiro que aponta para 'p1'
18 p1: .word value        # Primeiro ponteiro que aponta para 'valor'
19 value: .word 2
20
21 .text
22 .globl main
23 main:
24     ori $t0, $zero, 0x1001 # t0 = 0x00001001
25     sll $t0, $t0, 16 # t0 = 0x10010000
26
27     lw $t1, 0 ($t0) # t1 = p2
28     lw $t2, 0 ($t1) # t2 = p1
29     lw $t3, 0 ($t2) # t3 = value
30     lw $s0, 0 ($t3) # k = 2
31
32     add $s0, $s0, $s0 # k = 2*k
33
34     sw $s0, 0 ($t3) # memoria[$t3] = k
35 #fim
```

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File Edit Run Settings Tools Help

Run speed at max (no interaction)

Edit Execute

Text Segment

Block	Address	Code	Basic	Source
0x00400000	0x34081001	ori \$8,\$0,0x00001001	24:	ori \$t0,\$zero,0x1001 # t0 = 0x00001001
0x00400004	0x00084400	sll \$8,\$8,0x00000010	25:	sll \$t0,\$t0,16 # t0 = 0x10010000
0x00400008	0x8d950000	lw \$9,0x00000000(\$8)	27:	lw \$t1,0(\$t0) # t1 = p2
0x0040000c	0x8d2a0000	lw \$10,0x00000000(\$9)	28:	lw \$t2,0(\$t1) # t2 = p1
0x00400010	0x8d450000	lw \$11,0x00000000(\$...	29:	lw \$t3,0(\$t2) # t3 = value
0x00400014	0xd4700000	lw \$16,0x00000000(\$...	30:	lw \$a0,0(\$t3) # k = 2
0x00400018	0x02108020	add \$16,\$16,\$16	32:	add \$a0,\$a0,\$a0 # k = 2*k
0x0040001c	0xad700000	sw \$16,0x00000000(\$...	34:	sw \$a0,0(\$t3) # memoria[\$t3] = k

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x10010004	0x10010008	0x1001000c	0x00000004	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Mars Messages Run I/O

-- program is finished running (dropped off bottom) --

Clear

-- program is finished running (dropped off bottom) --

Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x10010000
\$t1	9	0x10010004
\$t2	10	0x10010008
\$t3	11	0x1001000c
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00000002
\$s1	17	0x00000000
\$s2	18	0x00000000
\$s3	19	0x00000000
\$s4	20	0x00000000
\$s5	21	0x00000000
\$s6	22	0x00000000
\$s7	23	0x00000000
\$s8	24	0x00000000
\$t9	25	0x00000000
\$k0	26	0x00000000
\$k1	27	0x00000000
\$gp	28	0x10000000
\$fp	29	0x7ffffc00
\$ra	30	0x00000000
\$ra	31	0x00000000
pc		0x00400020
hi		0x00000000
lo		0x00000000

Programa 13:

Edit Execute

mips1.asm

```

1  #programa 13:
2  #Escreva um programa que leia um valor A da memória, identifique se o número é negativo ou
3  #não e encontre o seu módulo. O valor deverá ser reescrito sobre A.
4
5  # a = $a0;
6  #inicio
7  .data
8  A: .word -2
9
10 .text
11 .globl main
12 main:
13     ori $t0,$zero,0x1001 # t0 = 0x00001001
14     sll $t0,$t0,16 # t0 = 0x10010000
15
16     lw $a0,0($t0) # a = memoria[0]
17
18     sra $t1,$a0,31 # Desloca o bit de sinal para todos os bits para verificar se o número é negativo
19     beq $t1,$zero,ehPositivo
20     sub $a0,$zero,$a0
21
22     ehPositivo:
23         sw $a0,0($t0)
24 #fim

```

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File Edit Run Settings Tools Help

Run speed at max (no interaction)

Edit Execute

Text Segment

Block	Address	Code	Basic	Source
0x00400000	0x34081001	ori \$8,\$0,0x00001001	13:	ori \$t0,\$zero,0x1001 # t0 = 0x00001001
0x00400004	0x00084400	sll \$8,\$8,0x00000010	14:	sll \$t0,\$t0,16 # t0 = 0x10010000
0x00400008	0x8d100000	lw \$16,0x00000000(\$8)	16:	lw \$a0,0(\$t0) # a = memoria[0]
0x0040000c	0x00104fc3	sra \$8,\$8,0x0000001f	18:	sra \$t1,\$a0,31 # Desloca o bit de sinal para todos...
0x00400010	0x11200001	beq \$8,\$zero,ehPositivo	19:	beq \$t1,\$zero,ehPositivo
0x00400014	0x00108020	add \$16,\$16,\$16	20:	add \$a0,\$a0,\$a0
0x00400018	0xad100000	sw \$16,0x00000000(\$8)	23:	sw \$a0,0(\$t0)

Labels

Label	Address
(global)	
main	0x00400000
ehPositivo	0x00400018
A	0x10010000

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000002	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010020	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010040	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010060	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010080	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100a0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100c0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x100100e0	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010100	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010120	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Mars Messages Run I/O

-- program is finished running (dropped off bottom) --

Clear

-- program is finished running (dropped off bottom) --

Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$t0	8	0x10010000
\$t1	9	0xffffffff
\$t2	10	0x00000000
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$s0	16	0x00000002
\$s1	17	0x00000000
\$s2	18	0x00000000
\$s3	19	0x00000000
\$s4	20	0x00000000
\$s5	21	0x00000000
\$s6	22	0x00000000
\$s7	23	0x00000000
\$s8	24	0x00000000
\$t9	25	0x00000000
\$k0	26	0x00000000
\$k1	27	0x00000000
\$gp	28	0x10000000
\$fp	29	0x7ffffc00
\$ra	30	0x00000000
\$ra	31	0x00000000
pc		0x0040001e
hi		0x00000000
lo		0x00000000

Programa 14:

EditExecute

mips1.asm

```
1 #programa 14:
2 #Escreva um programa que leia um valor A da memória, identifique se o número é par ou não.
3 #Um valor deverá ser escrito na segunda posição livre da memória (0 para par e 1 para ímpar).
4
5 # a = $s0;
6 #início
7 .data
8 A: .word 3
9
10 .text
11 .globl main
12 main:
13 ori $t0, $zero, 0x1001 # t0 = 0x00001001
14 sll $t0, $t0, 16 # t0 = 0x10010000
15
16 lw $a0, 0 ($t0) # a = memoria[0]
17
18 andi $t1, $a0, 1 # para ver se o ultimo bit e 0 ou 1
19 beq $t1, 0, par
20 ori $t1, $zero, 0
21 j fim
22
23 par:
24 ori $t1, $zero, 1
25
26 fim:
27 sw $t1, 4($t0)
```

FileEditRunSettingsToolsHelp

Run speed at max (no interaction)

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x34081000	ori \$t0, \$zero, 0x1001	13: ori \$t0, \$zero, 0x1001 # t0 = 0x00001001
	0x00400004	0x00084000	sll \$t0, \$t0, 16	14: sll \$t0, \$t0, 16 # t0 = 0x10010000
	0x00400008	0x8d100000	lw \$a0, 0(\$t0)	16: lw \$a0, 0 (\$t0) # a = memoria[0]
	0x0040000c	0x32090001	andi \$t1, \$a0, 1	18: andi \$t1, \$a0, 1 # para ver se o ultimo bit e 0 ou 1
	0x00400010	0x20010000	beq \$t1, 0, par	19: beq \$t1, 0, par
	0x00400014	0x10290002	ori \$t1, \$zero, 0	20: ori \$t1, \$zero, 0
	0x00400018	0x34090000	j fim	21: j fim
	0x0040001c	0x08100009	ori \$t1, \$zero, 1	24: ori \$t1, \$zero, 1
	0x00400020	0x34090001	sw \$t1, 4(\$t0)	27: sw \$t1, 4(\$t0)

Labels

Label	Address
(global)	
main	0x00400000
par	0x00400020
fim	0x00400024
A	0x10010000

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	3	0	0	0	0	0	0	0
0x10010004	0	0	0	0	0	0	0	0
0x10010008	0	0	0	0	0	0	0	0
0x1001000c	0	0	0	0	0	0	0	0
0x10010010	0	0	0	0	0	0	0	0
0x10010014	0	0	0	0	0	0	0	0
0x10010018	0	0	0	0	0	0	0	0
0x1001001c	0	0	0	0	0	0	0	0
0x10010020	0	0	0	0	0	0	0	0

Registers

Name	Number	Value
\$zero	0	0
\$at	1	0
\$v0	2	0
\$v1	3	0
\$a0	4	0
\$a1	5	0
\$a2	6	0
\$a3	7	0
\$t0	8	268500992
\$t1	9	1
\$t2	10	0
\$t3	11	0
\$t4	12	0
\$t5	13	0
\$t6	14	0
\$t7	15	0
\$a0	16	3
\$a1	17	0
\$a2	18	0
\$a3	19	0
\$a4	20	0
\$a5	21	0
\$a6	22	0
\$a7	23	0
\$t8	24	0
\$t9	25	0
\$t0	26	0
\$t1	27	0
\$gp	28	268468224
\$fp	29	2147475548
\$sp	30	0
\$ra	31	0
\$pc		4194344
\$hi		0
\$lo		0

Mars Messages

Run I/O

Clear

-- program is finished running (dropped off bottom) --

-- program is finished running (dropped off bottom) --

Programa 15:

EditExecute

mips1.asm

```

4  #Use o MARS para verificar a quantidade de instruções conforme o tipo (ULA, Desvios, Mem ou Outras)
5
6  .text
7  .globl main
8  main:
9      ori $t0, $zero, 0x1001 # t0 = 0x00001001
10     sll $t0, $t0, 16 # t0 = 0x10010000
11
12     ori $s0, $zero, 0 # i = 0
13     ori $s1, $zero, 0 # soma = 0
14     ori $s2, $zero, 0 # result = 0
15     ori $t1, $zero, 100 # n = 100
16
17     loop:
18         sll $t2, $s0, 1 # 2*i
19         addi $s2, $t2, 1 # result = t2 + 1
20         sw $s2, 0($t0) # memoria[i] = result
21
22         add $s1, $s1, $s2 # soma += result
23
24         addi $t0, $t0, 4 # pos += 4 -> próximo endereço
25
26         addi $s0, $s0, 1 # i++
27         bne $s0, $t1, loop # if(i != 100)
28
29     sw $s1, 0($t0) # memoria[i] = soma -> grava soma após o último elemento
30     #fim

```

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File Edit Run Settings Tools Help

Run speed at max (no interaction)

EditExecute

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	ori	\$t0, \$zero, 0x1001	ori \$t0, \$zero, 0x1001 # t0 = 0x00001001
	0x00400004	sll	\$t0, \$t0, 16	sll \$t0, \$t0, 16 # t0 = 0x10010000
	0x00400008	ori	\$s0, \$zero, 0	ori \$s0, \$zero, 0 # i = 0
	0x0040000c	ori	\$s1, \$zero, 0	ori \$s1, \$zero, 0 # soma = 0
	0x00400010	ori	\$s2, \$zero, 0	ori \$s2, \$zero, 0 # result = 0
	0x00400014	ori	\$t1, \$zero, 100	ori \$t1, \$zero, 100 # n = 100
	0x00400018	sll	\$t2, \$s0, 1	sll \$t2, \$s0, 1 # 2*i
	0x0040001c	addi	\$s2, \$t2, 1	addi \$s2, \$t2, 1 # result = t2 + 1
	0x00400020	sw	\$s2, 0(\$t0)	sw \$s2, 0(\$t0) # memoria[i] = result
	0x00400024	add	\$s1, \$s1, \$s2	add \$s1, \$s1, \$s2 # soma += result
	0x00400028	addi	\$t0, \$t0, 4	addi \$t0, \$t0, 4 # pos += 4 -> próximo endereço
	0x0040002c	addi	\$s0, \$s0, 1	addi \$s0, \$s0, 1 # i++

Labels

Label	Address
(global)	
main	0x00400000
mips1.asm	
loop	0x00400018

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	97	99	101	103	105	107	109	111
0x10010004	113	115	117	119	121	123	125	127
0x10010008	129	131	133	135	137	139	141	143
0x1001000c	145	147	149	151	153	155	157	159
0x10010010	161	163	165	167	169	171	173	175
0x10010014	177	179	181	183	185	187	189	191
0x10010018	193	195	197	199	10000	0	0	0
0x1001001c	0	0	0	0	0	0	0	0
0x10010020	0	0	0	0	0	0	0	0
0x10010024	0	0	0	0	0	0	0	0

Mars Messages

Run I/O

Clear

-- program is finished running (dropped off bottom) --

-- program is finished running (dropped off bottom) --

Registers

Name	Coproc 1	Coproc 0	Value
\$zero			0
\$at			1
\$v0			2
\$v1			3
\$a0			4
\$a1			5
\$a2			6
\$a3			7
\$t0			8
\$t1			9
\$t2			10
\$t3			11
\$t4			12
\$t5			13
\$t6			14
\$t7			15
\$s0			16
\$s1			17
\$s2			18
\$s3			19
\$s4			20
\$s5			21
\$s6			22
\$s7			23
\$s8			24
\$s9			25
\$k0			26
\$k1			27
\$gp			28
\$sp			29
\$fp			30
\$ra			31
\$pc			4194360
\$hi			0
\$lo			0

Programa 16:

EditExecute

mips2.asm

mips1.asm

```

13
14 .text
15 .globl main
16 main:
17     ori $t0, $zero, 0x1001 # t0 = 0x00001001
18     sll $t0, $t0, 16 # t0 = 0x10010000
19
20     lw $a0, 0 ($t0) # x = 0x186A00
21     srl $t1, $a0, 4 # t1 = 0x186A0
22
23     lw $a1, 4 ($t0) # y = 0x13880
24     srl $t2, $a1, 4 # t2 = 0x1388
25
26     lw $a2, 8 ($t0) # z = 0x61A80
27     srl $t3, $a2, 4 # t3 = 0x61A8
28
29     mult $t1, $t2 # x * y
30     mflo $t4 # t4 = x * y
31
32     div $t4, $t3 # t4 / t3
33     mflo $t5 # t5 = t4 / t3
34
35     sll $t5, $t5, 4 # vai mover os 4bits para a esquerda que eu tirei
36     sw $t5, 12 ($t0) # mempria[3] = t5
37
38     #fim
39

```

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FileEditRunSettingsToolsHelp

Run speed at max (no interaction)

EditExecute

Text Segment

Offset	Address	Code	Basic	Source
0x00400000	0x34981001	ori	\$t0, \$zero, 0x1001	17: ori \$t0, \$zero, 0x1001 # t0 = 0x00001001
0x00400004	0x00084400	sll	\$t0, \$t0, 16	18: sll \$t0, \$t0, 16 # t0 = 0x10010000
0x00400008	0x8d100000	lw	\$a0, 0 (\$t0)	20: lw \$a0, 0 (\$t0) # x = 0x186A00
0x0040000c	0x00104902	srl	\$t1, \$a0, 4	21: srl \$t1, \$a0, 4 # t1 = 0x186A0
0x00400010	0x8d110004	lw	\$a1, 4 (\$t0)	23: lw \$a1, 4 (\$t0) # y = 0x13880
0x00400014	0x00115102	srl	\$t2, \$a1, 4	24: srl \$t2, \$a1, 4 # t2 = 0x1388
0x00400018	0x8d120008	lw	\$a2, 8 (\$t0)	26: lw \$a2, 8 (\$t0) # z = 0x61A80
0x0040001c	0x00125902	srl	\$t3, \$a2, 4	27: srl \$t3, \$a2, 4 # t3 = 0x61A8
0x00400020	0x012a0018	mult	\$t1, \$t2	29: mult \$t1, \$t2 # x * y
0x00400024	0x00006012	mflo	\$t4	30: mflo \$t4 # t4 = x * y
0x00400028	0x018b001a	div	\$t4, \$t3	32: div \$t4, \$t3 # t4 / t3
0x0040002c	0x00006812	mflo	\$t5	33: mflo \$t5 # t5 = t4 / t3

Labels

Label	Address
(global)	
main	0x00400000
x	0x10010000
y	0x10010004
z	0x10010008

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	1600000	80000	400000	320000	0	0	0	0
0x10010020	0	0	0	0	0	0	0	0
0x10010040	0	0	0	0	0	0	0	0
0x10010060	0	0	0	0	0	0	0	0
0x10010080	0	0	0	0	0	0	0	0
0x100100a0	0	0	0	0	0	0	0	0
0x100100c0	0	0	0	0	0	0	0	0
0x100100e0	0	0	0	0	0	0	0	0
0x10010100	0	0	0	0	0	0	0	0
0x10010120	0	0	0	0	0	0	0	0

Registers

Name	Coproc 1	Coproc 0	Value
\$zero			0
\$at			0
\$v0			0
\$v1			0
\$a0			0
\$a1			0
\$a2			0
\$a3			0
\$t0			268500592
\$t1			100000
\$t2			50000
\$t3			25000
\$t4			500000000
\$t5			320000
\$t6			0
\$t7			0
\$a0			1600000
\$a1			80000
\$a2			400000
\$a3			0
\$a4			0
\$a5			0
\$a6			0
\$a7			0
\$a8			0
\$a9			0
\$a10			0
\$a11			0
\$a12			0
\$a13			0
\$a14			0
\$a15			0
\$a16			0
\$a17			0
\$a18			0
\$a19			0
\$a20			0
\$a21			0
\$a22			0
\$a23			0
\$a24			0
\$a25			0
\$a26			0
\$a27			0
\$a28			0
\$a29			0
\$a30			0
\$a31			0
\$a32			0
\$a33			0
\$a34			0
\$a35			0
\$a36			0
\$a37			0
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\$a81			0
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\$a83			0
\$a84			0
\$a85			0
\$a86			0
\$a87			0
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\$a89			0
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\$a91			0
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\$a93			0
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\$a126			0
\$a127			0
\$a128			0
\$a129			0
\$a130			0
\$a131			0
\$a132			0
\$a133			0
\$a134			0
\$a135			0
\$a136			0
\$a137			0
\$a138			0
\$a139			0
\$a140			0
\$a141			0
\$a142			0
\$a143			0
\$a144			0
\$a145			0
\$a146			0
\$a147			0
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\$a170			0
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\$a173			0
\$a174			0
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\$a184			0
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\$a224			0
\$a225			0
\$a226			0
\$a227			0
\$a228			0
\$a229			0
\$a230			0
\$a231			0
\$a232			0
\$a233			0
\$a234			0
\$a235			0
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\$a237			0
\$a238			0
\$a239			0
\$a240			0
\$a241			0
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\$a245			0
\$a246			0
\$a247			0
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\$a249			0
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\$a251			0
\$a252			0
\$a253			0
\$a254			0
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\$a258			0
\$a259			0
\$a260			0
\$a261			0
\$a262			0
\$a263			0
\$a264			0
\$a265			0
\$a266			0

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File Edit Run Settings Tools Help

Run speed at max (no interaction)

Edit Execute

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x34981001	ori \$t0, \$zero, 0x1001 # t0 = 0x00001001	18: ori \$t0, \$zero, 0x1001 # t0 = 0x00001001
	0x00400004	0x00084400	sll \$t0, \$t0, 16 # t0 = 0x10010000	19: sll \$t0, \$t0, 16 # t0 = 0x10010000
	0x00400008	0x8d100001	lw \$a0, 0 (\$t0) # x = memoria[0]	21: lw \$a0, 0 (\$t0) # x = memoria[0]
	0x0040000c	0x8d110004	lw \$a1, 4 (\$t0) # y = memoria[1]	22: lw \$a1, 4 (\$t0) # y = memoria[1]
	0x00400010	0x34090000	ori \$t1, \$zero, 0 # i = 0	24: ori \$t1, \$zero, 0 # i = 0
	0x00400014	0x340a0000	ori \$t2, \$zero, 0 # soma = 0	25: ori \$t2, \$zero, 0 # soma = 0
	0x00400018	0x01505020	add \$t0, \$t2, \$a0 # soma += x	28: add \$t0, \$t2, \$a0 # soma += x
	0x0040001c	0x21290001	addi \$t1, \$t1, 1 # i++	29: addi \$t1, \$t1, 1 # i++
	0x00400020	0x1531ffff	bne \$t1, \$a1, loop # if(i != y)	30: bne \$t1, \$a1, loop # if(i != y)
	0x00400024	0x000a5020	add \$a2, \$zero, \$t2 # k = soma	32: add \$a2, \$zero, \$t2 # k = soma
	0x00400028	0xad120008	sw \$a2, 8 (\$t0) # memoria[2] = k	33: sw \$a2, 8 (\$t0) # memoria[2] = k

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	0x00000002	0x00000003	0x00000006	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010002	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010004	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010006	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010008	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001000a	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001000c	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x1001000e	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010010	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000
0x10010012	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000	0x00000000

Mars Messages Run I/O

Clear

program is finished running (dropped off bottom) --

program is finished running (dropped off bottom) --

Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0x00000000
\$at	1	0x00000000
\$v0	2	0x00000000
\$v1	3	0x00000000
\$a0	4	0x00000000
\$a1	5	0x00000000
\$a2	6	0x00000000
\$a3	7	0x00000000
\$a0	8	0x10010000
\$t1	9	0x00000003
\$t2	10	0x00000006
\$t3	11	0x00000000
\$t4	12	0x00000000
\$t5	13	0x00000000
\$t6	14	0x00000000
\$t7	15	0x00000000
\$a0	16	0x00000002
\$a1	17	0x00000003
\$a2	18	0x00000006
\$a3	19	0x00000000
\$a4	20	0x00000000
\$a5	21	0x00000000
\$a6	22	0x00000000
\$a7	23	0x00000000
\$t8	24	0x00000000
\$t9	25	0x00000000
\$t0	26	0x00000000
\$t1	27	0x00000000
\$gp	28	0x10008000
\$gp	29	0x7ffffc
\$fp	30	0x00000000
\$ra	31	0x00000000
\$pc		0x0040002c
\$hi		0x00000000
\$lo		0x00000000

Programa 18:

Edit Execute

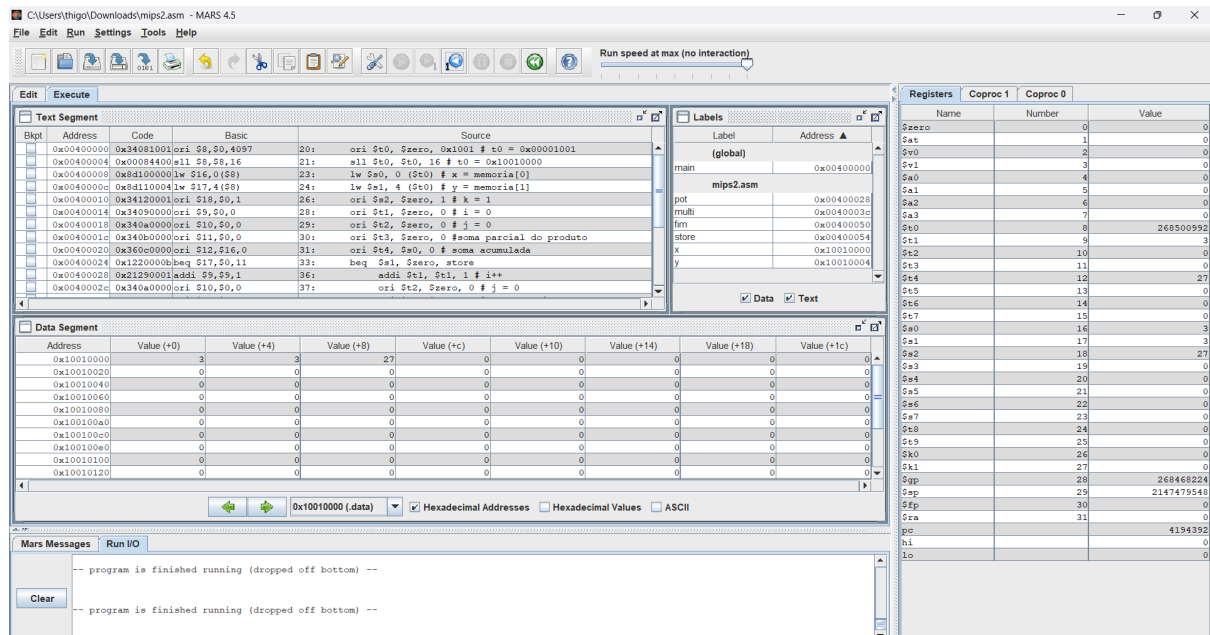
mips2.asm*

```

19 main:
20     ori $t0, $zero, 0x1001 # t0 = 0x00001001
21     sll $t0, $t0, 16 # t0 = 0x10010000
22     lw $a0, 0 ($t0) # x = memoria[0]
23     lw $a1, 4 ($t0) # y = memoria[1]
24     ori $s2, $zero, 1 # k = 1
25     ori $t1, $zero, 0 # i = 0
26     ori $t2, $zero, 0 # j = 0
27     ori $t3, $zero, 0 # soma parcial do produto
28     ori $t4, $a0, 0 # soma acumulada
29     beq $a1, $zero, store
30
31     pot:
32         addi $t1, $t1, 1 # i++
33         ori $t2, $zero, 0 # j = 0
34         ori $t3, $zero, 0 # soma parcial = 0
35         bne $t1, $a1, multi # if(i != y)
36         j fim
37     multi:
38         add $t3, $t3, $t4 # soma parcial += t4
39         addi $t2, $t2, 1 # j++
40         bne $t2, $a0, multi # if(j != x)
41         ori $t4, $t3, 0 # soma acumulada = soma parcial
42         j pot
43     fim:
44         add $s2, $zero, $t4 # k = soma acumulada
45     store:
46         sw $s2, 8 ($t0) # memoria[2] = k

```

Line: 22 Column: 1 ☒ Show Line Numbers



Perguntas:

1. Se tivermos 2 inteiros, cada um com 32 bits, quantos bits podemos esperar para o produto?

C. 64

2. Quais os registradores que armazenam os resultados na multiplicação?

B. hi e lo

3. Qual a operação usada para multiplicar inteiros em comp. de dois?

A. mult

4. Qual instrução move os bits menos significativos da multiplicação para o reg. 8?

C. mflo \$8

5. Se tivermos dois inteiros, cada um com 32 bits, quantos bits deveremos estar preparados para receber no quociente?

B. 32

6. Após a instrução div, qual registrador possui o quociente?

A. lo

7. Qual a inst. Usada para dividir dois inteiros em comp. de dois?

D. div

8. Faça um arithmetic shift right de dois no seguinte padrão de bits: 1001 1011

A. 1110 0110

9. Qual o efeito de um arithmetic shift right de uma posição?

C. Se o inteiro for unsigned, o shift pode ocasionar um valor errado. Se o inteiro for signed, o shift o divide por 2.

10. Qual sequencia de instruções avalia $3x+7$, onde x é iniciado no reg. \$8 e o resultado armazenado em \$9?

A.

ori \$3,\$0,3

mult \$8,\$3

mflo \$9

addi \$9,\$9,7

Implemente os programas a seguir no MIPS/MARS:

Programa 19:

```
15  main:
16      ori $t2, $zero, 0x1001 # t0 = 0x00001001
17      sll $t2, $t2, 16 # t0 = 0x10010000
18
19      lw $s0, 0 ($t2) # s0 = memoria[0]
20      lw $s1, 4 ($t2) # s1 = memoria[1]
21      ori $s2, $zero, 0 # resp = 0
22
23      beq $s0, $zero, store2 # caso zero
24      beq $s1, $zero, store2 # caso zero
25
26      addi $t0, $zero, 0 # t0 = 0
27      addi $t1, $zero, 0 # t1 = 0
28
29      or $t3, $s0, $zero # t3 = s0
30      or $t4, $s1, $zero # t4 = s1
31
32      cont1:
33          addi $t0, $t0, 1 # t0++
34          srl $t3, $t3, 1 # movimenta o t3 um bit para direita
35          bne $t3, $zero, cont1 # if(t3 != 0) -> volta para cont1
36
37      cont2:
38          addi $t1, $t1, 1 # t1++
39          srl $t4, $t4, 1 # movimenta o t4 um bit para direita
40          bne $t4, $zero, cont2 # if(t4 != 0) -> volta para cont2
41
```

```

42      check:
43          slti $t5, $t0, 32 # se t0 < 32 -> t5 = 1
44          slti $t6, $t1, 32 # se t1 < 32 -> t6 = 1
45          add $t7, $t5, $t6 # t7 = t5 + t6
46
47          slti $t7, $t7, 2 # se t7 < 2, então t7 = 1
48          beq $t7, $zero, multi2 # if(t7 == 0) -> vai para multi2
49
50      multi1:
51          mult $s0, $s1 # s0 * s1
52          mflo $s2 # s2 recebe os bits menos significativos de resultado
53          mfhi $s3 # s3 recebe os bits mais significativos de resultado
54          j store1
55
56      multi2:
57          mult $s0, $s1 # s0 * s1
58          mflo $s2 # s2 recebe os bits menos significativos de resultado
59          j store2
60
61      store1:
62          sw $s2, 8 ($t2) # memoria[2] = s2
63          sw $s3, 12 ($t2) # memoria[3] = s3
64          j fim
65
66      store2:
67          sw $s2, 8 ($t2) # memoria[2] = s2
68
69      fim:
70
71      #fim

```

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Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0
\$at	1	0
\$v0	2	0
\$v1	3	0
\$a0	4	0
\$a1	5	0
\$a2	6	0
\$a3	7	0
\$t0	8	4
\$t1	9	3
\$t2	10	268500992
\$t3	11	0
\$t4	12	0
\$t5	13	1
\$t6	14	1
\$t7	15	0
\$s0	16	8
\$s1	17	7
\$s2	18	56
\$s3	19	0
\$s4	20	0
\$s5	21	0
\$s6	22	0
\$s7	23	0
\$s8	24	0
\$s9	25	0
\$k0	26	0
\$k1	27	0
\$fp	28	268468224
\$gp	29	2147479548
\$tp	30	0
\$ra	31	0
pc		4194424
hi		0
lo		56

Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x340a1001	ori \$t0,\$zero,0x1001	ori \$t0, \$zero, 0x1001 # t0 = 0x00001001
	0x00400004	0x000a5400	sll \$t2,\$t2,16	sll \$t2, \$t2, 16 # t2 = 0x10010000
	0x00400008	0x8d500000	lw \$s0,0(\$t2)	lw \$s0, 0 (\$t2) # s0 = memoria[0]
	0x0040000c	0x8d510004	lw \$s1,4(\$t2)	lw \$s1, 4 (\$t2) # s1 = memoria[1]
	0x00400010	0x20080000	addi \$t0,\$zero,0	addi \$t0, \$zero, 0 # t0 = 0
	0x00400014	0x20090000	addi \$t1,\$zero,0	addi \$t1, \$zero, 0 # t1 = 0
	0x00400018	0x02005825	or \$t3,\$s0,\$s0	or \$t3, \$s0, \$s0 # t3 = s0
	0x0040001c	0x02206025	or \$t4,\$s1,\$zero	or \$t4, \$s1, \$zero # t4 = s1
	0x00400020	0x21080001	addi \$t0,\$t0,1	addi \$t0, \$t0, 1 # t0++
	0x00400024	0x000b5842	srl \$t3,\$t3,1	srl \$t3, \$t3, 1 # movimenta o t3 um bit para direita
	0x00400028	0x1560fffb	bne \$t3,\$zero,cont1	bne \$t3, \$zero, cont1 # if(t3 != 0) -> volta para cont1
	0x0040002c	0x21290001	addi \$t1,\$t1,1	addi \$t1, \$t1, 1 # t1++

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	8	7	56	0	0	0	0	0
0x10010020	0	0	0	0	0	0	0	0
0x10010040	0	0	0	0	0	0	0	0
0x10010060	0	0	0	0	0	0	0	0
0x10010080	0	0	0	0	0	0	0	0
0x100100a0	0	0	0	0	0	0	0	0
0x100100c0	0	0	0	0	0	0	0	0
0x100100e0	0	0	0	0	0	0	0	0
0x10010100	0	0	0	0	0	0	0	0
0x10010120	0	0	0	0	0	0	0	0

Mars Messages Run I/O

-- program is finished running (dropped off bottom) --

Clear

-- program is finished running (dropped off bottom) --

Programa 20:

mips2.asm*

```

7
8 # x= $s0; y= $s1;
9 .data
10     x: .word 3
11
12 #inicio
13 .text
14 .globl main
15 main:
16     ori $t0, $zero, 0x1001 # t0 = 0x00001001
17     sll $t0, $t0, 16 # t0 = 0x10010000
18     lw $s0, 0 ($t0) # x = memoria[0]
19     andi $t1, $s0, 1 # ver se o ultimo bit é 0 ou 1
20     add $t2, $zero, $s0 # t2 = x
21     mult $t2, $t2 # t2 = x^2
22     mflo $t2 # t2 = x^2
23     beq $t1, $zero, ehPar #se t1 for par manda para o label ehPar
24     ehImpar:
25         mult $t2, $s0 # x^3
26         mflo $t2 # t2 = x^3
27         sub $t3, $zero, $t2 # t3 = -x^3
28         mult $t2, $s0 # x^4
29         mflo $t2 # t2 = x^4
30         mult $t2, $s0 # x^5
31         mflo $t2 # t2 = x^5
32         add $t3, $t3, $t2 #
33         add $t3, $t3, 1 # t3 += 1
34     j fim
35     ehPar:
36         add $t3, $t2, $t2 # t3 = 2*t2
37         sub $t3, $zero, $t3 # t3 = -2*t2
38         mult $t2, $s0 # x^3
39         mflo $t2 # t2 = x^3
40         add $t3, $t2, $t3 # t3 += t2
41         mult $t2, $s0 # x^4
42         mflo $t2 # t2 = x^4
43         add $t3, $t2, $t3 # t3 += t2
44     fim:
45         add $s1, $zero, $t3 # y = t3
46         sw $s1, 4 ($t0) # memoria[1] = y
47 #fim
48

```

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Text Segment

Bkpt	Address	Code	Basic	Source
	0x00400000	0x34081001	ori \$t0, \$zero, 0x1001	16: ori \$t0, \$zero, 0x1001 # t0 = 0x00001001
	0x00400004	0x00084400	sll \$t0, \$t0, 16	17: sll \$t0, \$t0, 16 # t0 = 0x10010000
	0x00400008	0x8d100000	lw \$s0, 0 (\$t0)	18: lw \$s0, 0 (\$t0) # x = memoria[0]
	0x0040000c	0x2090001e	andi \$t1, \$s0, 1	19: andi \$t1, \$s0, 1 # ver se o ultimo bit é 0 ou 1
	0x00400010	0x00105020	add \$t2, \$zero, \$s0	20: add \$t2, \$zero, \$s0 # t2 = x
	0x00400014	0x014a0018	mult \$t2, \$t2	21: mult \$t2, \$t2 # t2 = x^2
	0x00400018	0x00005012	mflo \$t2	22: mflo \$t2 # t2 = x^2
	0x0040001c	0x1120000a	beq \$t1, \$zero, ehPar	23: beq \$t1, \$zero, ehPar #se t1 for par manda para o label ehPar
	0x00400020	0x01500018	mult \$t2, \$s0	24: mult \$t2, \$s0 # t2 = x^3
	0x00400024	0x00005012	mflo \$t2	25: mflo \$t2 # t2 = x^3
	0x00400028	0x000a5822	sub \$t3, \$zero, \$t2	26: sub \$t3, \$zero, \$t2 # t3 = -x^3
	0x0040002c	0x01500018	mult \$t2, \$s0	27: mult \$t2, \$s0 # t2 = x^4

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+c)	Value (+10)	Value (+14)	Value (+18)	Value (+1c)
0x10010000	3	217	0	0	0	0	0	0
0x10010020	0	0	0	0	0	0	0	0
0x10010040	0	0	0	0	0	0	0	0
0x10010060	0	0	0	0	0	0	0	0
0x10010080	0	0	0	0	0	0	0	0
0x100100a0	0	0	0	0	0	0	0	0
0x100100c0	0	0	0	0	0	0	0	0
0x100100e0	0	0	0	0	0	0	0	0
0x10010100	0	0	0	0	0	0	0	0
0x10010120	0	0	0	0	0	0	0	0

Mars Messages

Run I/O

Clear

-- program is finished running (dropped off bottom) --

-- program is finished running (dropped off bottom) --

Registers

Name	Number	Value
\$zero	0	0
\$at	1	0
\$v0	2	0
\$v1	3	0
\$a0	4	0
\$a1	5	0
\$a2	6	0
\$a3	7	0
\$t0	8	268500992
\$t1	9	1
\$t2	10	243
\$t3	11	217
\$t4	12	0
\$t5	13	0
\$t6	14	0
\$t7	15	0
\$s0	16	3
\$s1	17	217
\$s2	18	0
\$s3	19	0
\$s4	20	0
\$s5	21	0
\$s6	22	0
\$s7	23	0
\$s8	24	0
\$s9	25	0
\$k0	26	0
\$k1	27	0
\$gp	28	268468224
\$fp	29	2147479540
\$fp	30	0
\$ra	31	0
pc		4194416
hi		0
lo		243

Programa 21:

```

mips2.asm*
7  # x= $s0; y= $s1;
8  .data
9      x: .word -2
10
11 #inicio
12 .text
13 .globl main
14 main:
15     ori $t0, $zero, 0x1001 # t0 = 0x00001001
16     sll $t0, $t0, 16 # t0 = 0x10010000
17     lw $s0, 0 ($t0) # x = memoria[0]
18     andi $t1, $s0, 1 # ver se o ultimo bit é 0 ou 1
19     add $t2, $zero, $s0 # t2 = x
20     mult $t2, $t2 # x^2
21     mflo $t2 # t2 = x^2
22     mult $t2, $s0 # x^3
23     mflo $t2 # t2 = x^3
24     slt $t1, $zero, $s0 # t1 = (0 < s0) ? 1 : 0
25     beq $t1, 1, maior # if(t1 == 1) -> manda para o label 'maior'
26     menorOuIgual:
27         subi $t3, $zero, 1 # t3 = -1
28         mult $t2, $s0 # x^4
29         mflo $t2 # t2 = x^4
30         add $t3, $t3, $t2 # t3 += t2
31         j fim
32     maior:
33         addi $t3, $t2, 1 # t3 = x^3 + 1
34
35     fim:
36         add $s1, $zero, $t3 # y = t3
37         sw $s1, 4 ($t0) # memoria[1] = y
38     #fim

```

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File Edit Run Settings Tools Help

Run speed at max (no interaction)

Registers Coproc 1 Coproc 0

Name	Number	Value
\$zero	0	0
\$at	1	1
\$v0	2	0
\$v1	3	0
\$a0	4	0
\$a1	5	0
\$a2	6	0
\$a3	7	0
\$t0	8	268500592
\$t1	9	0
\$t2	10	16
\$t3	11	15
\$t4	12	0
\$t5	13	0
\$t6	14	0
\$t7	15	0
\$a0	16	-2
\$a1	17	15
\$a2	18	0
\$a3	19	0
\$a4	20	0
\$a5	21	0
\$a6	22	0
\$a7	23	0
\$s0	24	0
\$s1	25	0
\$k0	26	0
\$k1	27	0
\$gp	28	26846224
\$sp	29	214747548
\$fp	30	0
\$ra	31	0
pc		4194388
\$1		0
lo		16

Mars Messages Run I/O

-- program is finished running (dropped off bottom) --

Clear

-- program is finished running (dropped off bottom) --